

Eugene Knowledge Graph

Production Cypher & GDS Query Catalog

June 20, 2026

Neo4j	5.26.9 (community)
APOC	5.26.9
GDS	2.13.5
Nodes	129375
Relationships	4050064
Queries executed	68
Captured live	June 20, 2026

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1. Schema & Metadata

Introspect the live graph model: labels, relationship types, property keys, counts and the APOC meta-graph.

1. Query: Total node count in the graph

Cypher:

```
MATCH (n)
RETURN count(n) AS total_nodes
```

Returned Result:

total_nodes

129375

1 row(s) · 55 ms

2. Query: Total relationship count in the graph

Cypher:

```
MATCH ()-[r]->()
RETURN count(r) AS total_relationships
```

Returned Result:

total_relationships

4050064

1 row(s) · 51 ms

3. Query: All node labels with per-label counts (descending)

Cypher:

```
MATCH (n)
UNWIND labels(n) AS label
RETURN label, count(*) AS cnt
ORDER BY cnt DESC
```

Returned Result:

label	cnt
biological_process	28642
gene_protein	27671
disease	17080
effect_phenotype	15311
anatomy	14035
molecular_function	11169
drug	7957
cellular_component	4176
...	...

9 row(s) · 248 ms

4. Query: All relationship types with per-type counts (top 12)

Cypher:

```
MATCH ()-[r]->()
RETURN type(r) AS rel_type, count(*) AS cnt
ORDER BY cnt DESC
LIMIT 12
```

Returned Result:

rel_type	cnt
anatomy_protein_present	1518203
drug_drug	1336314
protein_protein	321075
disease_phenotype_positive	150317
bioprocess_protein	144805
cellcomp_protein	83402
disease_protein	80411
molfunc_protein	69530
...	...

9 row(s) · 2867 ms

5. Query: Distinct property keys present in the database

Cypher:

```
CALL db.propertyKeys()
YIELD propertyKey
RETURN collect(propertyKey) AS property_keys
```

Returned Result:

property_keys
[node_id, node_name, node_source, node_index, display_relation, embed...

1 row(s) · 32 ms

6. Query: Whole-database meta statistics via APOC

Cypher:

```
CALL apoc.meta.stats()
YIELD labelCount, relTypeCount, nodeCount, relCount
RETURN labelCount, relTypeCount, nodeCount, relCount
```

Returned Result:

labelCount	relTypeCount	nodeCount	relCount
10	30	129375	4050064

1 row(s) · 52 ms

7. Query: Database components / versions (kernel, edition)

Cypher:

```
CALL dbms.components()
YIELD name, versions, edition
RETURN name, versions, edition
```

Returned Result:

name	versions	edition
Neo4j Kernel	[5.26.9]	community

1 row(s) · 29 ms

8. Query: Installed APOC and GDS plugin versions

Cypher:

```
RETURN apoc.version() AS apoc_version, gds.version() AS gds_version
```

Returned Result:

apoc_version	gds_version
5.26.9	2.13.5

1 row(s) · 23 ms

2. Retrieval & Filtering

Core read patterns: lookup by name, label scans, WHERE predicates, string matching, DISTINCT, ORDER / SKIP / LIMIT.

9. Query: Find a gene/protein node by exact name

Cypher:

```
MATCH (n:gene_protein {node_name: $name})
RETURN n.node_name, n.node_id, n.node_source
```

Returned Result:

n.node_name	n.node_id	n.node_source
UBC	7316	NCBI

1 row(s) · 50 ms

10. Query: List 5 drug nodes (label scan with LIMIT)

Cypher:

```
MATCH (d:drug)
RETURN d.node_name AS drug, d.node_index AS idx
LIMIT 5
```

Returned Result:

drug	idx
Copper	14012
Oxygen	14013
Flunisolide	14014
Alclometasone	14015
Medrysone	14016

5 row(s) · 31 ms

11. Query: Case-insensitive prefix search on disease names (STARTS WITH)

Cypher:

```
MATCH (d:disease)
WHERE toLower(d.node_name) STARTS
WITH 'anemia'
RETURN d.node_name AS disease
LIMIT 6
```

Returned Result:

disease
anemia of prematurity
anemia, congenital hypoplastic, with multiple congenital anomalies/in...
anemia due to chronic disorder
anemia, nonspherocytic hemolytic, possibly due to defect in porphyrin...
anemia, nonspherocytic hemolytic, associated with abnormality of red ...
anemia, nonspherocytic hemolytic, due to G6PD deficiency

6 row(s) · 51 ms

12. Query: Substring search on drug names (CONTAINS)

Cypher:

```
MATCH (d:drug)
WHERE d.node_name CONTAINS 'cillin'
RETURN d.node_name AS drug
LIMIT 6
```

Returned Result:

drug
Ampicillin
Nafcillin
Benzylpenicillin
Amoxicillin
Cloxacillin
Carbenicillin

6 row(s) · 38 ms

13. Query: Regular-expression match on gene names (=~)

Cypher:

```
MATCH (g:gene_protein)
WHERE g.node_name =~ '^TP5[0-9].*'
RETURN g.node_name AS gene
LIMIT 6
```

Returned Result:

gene
TP53
TP53AIP1
TP53BP1
TP53BP2
TP53COR1
TP53I11

6 row(s) · 172 ms

14. Query: Distinct node sources used across the graph

Cypher:

```
MATCH (n)
RETURN DISTINCT n.node_source AS source
ORDER BY source
```

Returned Result:

source
CTD
DrugBank
GO
HPO
MONDO
MONDO_grouped
NCBI

source
REACTOME
...

9 row(s) · 206 ms

15. Query: Pagination: diseases ordered by name, skip 10 take 5

Cypher:

```
MATCH (d:disease)
RETURN d.node_name AS disease
ORDER BY disease
SKIP 10
LIMIT 5
```

Returned Result:

disease
14q24.1q24.3 microdeletion syndrome
14q32 duplication syndrome
15q overgrowth syndrome
15q11q13 microduplication syndrome
15q14 microdeletion syndrome

5 row(s) · 51 ms

3. Aggregation & Grouping

count / avg / collect / percentile aggregations and grouped rollups.

16. Query: Average protein-protein degree across all proteins

Cypher:

```
MATCH (n:gene_protein)-[:protein_protein]-()
WITH n, count(*) AS deg
RETURN avg(deg) AS avg_degree, max(deg) AS max_degree, min(deg) AS min_degree
```

Returned Result:

avg_degree	max_degree	min_degree
34.98692	5197	1

1 row(s) · 358 ms

17. Query: Top 8 most-connected proteins by PPI degree (grouped count)

Cypher:

```
MATCH (n:gene_protein)-[:protein_protein]-()
RETURN n.node_name AS protein, count(*) AS degree
ORDER BY degree DESC
LIMIT 8
```

Returned Result:

protein	degree
UBC	5197
ETS1	1492
GATA2	1363
CTCF	1351
EP300	1117

protein	degree
MYC	1099
AR	1099
RAD21	961

8 row(s) · 455 ms

18. Query: Collect the first 10 drug names into a single list

Cypher:

```
MATCH (d:drug)
WITH d
LIMIT 10
RETURN collect(d.node_name) AS drug_sample
```

Returned Result:

drug_sample
[Copper, Oxygen, Flunisolide, Alclometasone, Medrysone, Fluorometholo...

1 row(s) · 34 ms

19. Query: Degree distribution percentiles for proteins (50/90/99th)

Cypher:

```
MATCH (n:gene_protein)-[:protein_protein]-()
WITH n, count(*) AS deg
RETURN percentileCont(deg,0.5) AS p50, percentileCont(deg,0.9) AS p90,
percentileCont(deg,0.99) AS p99
```

Returned Result:

p50	p90	p99
17	79	292.94

1 row(s) · 301 ms

20. Query: Relationship-type histogram for one hub protein

Cypher:

```
MATCH (n:gene_protein {node_name:$name})-[r]-()
RETURN type(r) AS rel_type, count(*) AS cnt
ORDER BY cnt DESC
```

Returned Result:

rel_type	cnt
protein_protein	5197
pathway_protein	176
anatomy_protein_present	134
bioprocess_protein	53
cellcomp_protein	14
molfunc_protein	5
drug_protein	1

7 row(s) · 54 ms

4. Pattern Matching & Traversal

One-hop neighbours, multi-hop expansion, variable-length paths, OPTIONAL MATCH and pattern comprehension.

21. Query: Direct protein interactors of a hub protein (one hop)

Cypher:

```
MATCH (n:gene_protein {node_name:$name})-[:protein_protein]-(m)
RETURN m.node_name AS interactor
LIMIT 6
```

Returned Result:

interactor
TMPO
TPM4
MCM3
CD247
CSNK1G2
DPYSL2

6 row(s) · 54 ms

22. Query: Proteins associated with a disease (typed one-hop)

Cypher:

```
MATCH (dis:disease)-[:disease_protein]-(p:gene_protein)
WHERE dis.node_name CONTAINS 'anemia'
RETURN dis.node_name AS disease, p.node_name AS protein
LIMIT 6
```

Returned Result:

disease	protein
iron deficiency anemia	TMPRSS6
iron deficiency anemia	TFRC
iron deficiency anemia	SOD1
iron deficiency anemia	SLC11A2
obsolete rare acquired hemolytic anemia	EIF2AK1
obsolete rare acquired hemolytic anemia	TPI1

6 row(s) · 176 ms

23. Query: Two-hop protein neighbourhood size (distinct)

Cypher:

```
MATCH (n:gene_protein {node_name:$name})-[:protein_protein*2]-(m)
RETURN count(DISTINCT m) AS two_hop_reach
```

Returned Result:

two_hop_reach
17305

1 row(s) · 374 ms

24. Query: Variable-length path 1..3 hops between two named proteins (existence)

Cypher:

```
MATCH (a:gene_protein {node_name:$src})
MATCH p = (a)-[:protein_protein*1..3]-(b:gene_protein {node_name:$dst})
RETURN length(p) AS hops
LIMIT 1
```

Returned Result:

hops

3

1 row(s) · 149 ms

25. Query: OPTIONAL MATCH: drug with its (possibly absent) indicated diseases

Cypher:

```
MATCH (d:drug {node_name:$drug})
OPTIONAL MATCH (d)-[:indication]-(x:disease)
RETURN d.node_name AS drug, collect(x.node_name)[0..5] AS indications
```

Returned Result:

drug	indications
Copper	[]

1 row(s) · 177 ms

26. Query: Pattern comprehension: a protein's interactor names inline

Cypher:

```
MATCH (n:gene_protein {node_name:$name})
RETURN [(n)-[:protein_protein]-(m) | m.node_name][0..6] AS interactors
```

Returned Result:

interactors
[S1PR1, CLK1, APLP2, CDK9, EIF4E, BCR]

1 row(s) · 176 ms

5. Path Finding

Native shortestPath / allShortestPaths plus GDS weighted Dijkstra.

27. Query: Shortest PPI path between two proteins (native shortestPath, ≤5 hops)

Cypher:

```
MATCH (a:gene_protein {node_name:$src}), (b:gene_protein {node_name:$dst}), p =
  shortestPath((a)-[:protein_protein*..5]-(b))
RETURN [x IN nodes(p) | x.node_name] AS path, length(p) AS hops
```

Returned Result:

path	hops
[UBC, ETS1]	1

1 row(s) · 190 ms

28. Query: Count of all shortest PPI paths between two proteins

Cypher:

```
MATCH (a:gene_protein {node_name:$src}), (b:gene_protein {node_name:$dst}), p =
  allShortestPaths((a)-[:protein_protein*..5]-(b))
RETURN count(p) AS num_shortest_paths, min(length(p)) AS hops
```

Returned Result:

num_shortest_paths	hops
1	1

1 row(s) · 103 ms

29. Query: Cross-domain reasoning path: drug → protein → disease (≤4 hops)

Cypher:

```
MATCH (d:drug {node_name:$drug}), (dis:disease)
WHERE dis.node_name CONTAINS 'anemia'
MATCH p = shortestPath((d)-[*..4]-(dis))
RETURN [x IN nodes(p) | coalesce(x.node_name, '?')] AS path, length(p) AS hops
LIMIT 1
```

Returned Result:

path	hops
[Copper, ALB, Sulfamethoxazole, folic acid deficiency anemia]	3

1 row(s) · 138 ms

6. GDS Graph Projection & Centrality

Project the protein-protein interaction (PPI) sub-network into the GDS in-memory catalog, then run the full suite of centrality algorithms. Centrality ranks the most structurally important proteins.

30. Query: Project the PPI network (gene_protein nodes + protein_protein edges, UNDIRECTED)

Cypher:

```
CALL gds.graph.project('ppi', 'gene_protein', {protein_protein: {orientation:
'UNDIRECTED'}})
YIELD graphName, nodeCount, relationshipCount
RETURN graphName, nodeCount, relationshipCount
```

Returned Result:

graphName	nodeCount	relationshipCount
ppi	27671	642150

1 row(s) · 576 ms

31. Query: Degree centrality — most-connected proteins (top 5)

Cypher:

```
CALL gds.degree.stream('ppi')
YIELD nodeId, score
RETURN gds.util.asNode(nodeId).node_name AS protein, score
ORDER BY score DESC
LIMIT 5
```

Returned Result:

protein	score
UBC	5197
ETS1	1492
GATA2	1363
CTCF	1351
EP300	1117

5 row(s) · 209 ms

32. Query: PageRank — influence by recursive importance (top 5)

Cypher:

```
CALL gds.pageRank.stream('ppi', {maxIterations:20, dampingFactor:0.85})
YIELD nodeId, score
RETURN gds.util.asNode(nodeId).node_name AS protein, score
ORDER BY score DESC
LIMIT 5
```

Returned Result:

protein	score
UBC	130.16015
CTCF	47.87065
GATA2	35.22993
ETS1	33.52944
EGR1	30.16246

5 row(s) · 539 ms

33. Query: ArticleRank — PageRank variant reducing low-degree bias (top 5)

Cypher:

```
CALL gds.articleRank.stream('ppi', {maxIterations:20})
YIELD nodeId, score
RETURN gds.util.asNode(nodeId).node_name AS protein, score
ORDER BY score DESC
LIMIT 5
```

Returned Result:

protein	score
UBC	21.8711
CTCF	6.14524
ETS1	6.11367
GATA2	5.74197
EP300	4.6981

5 row(s) · 516 ms

34. Query: Eigenvector centrality — influence weighted by neighbours' influence (top 5)

Cypher:

```
CALL gds.eigenvector.stream('ppi', {maxIterations:20})
YIELD nodeId, score
RETURN gds.util.asNode(nodeId).node_name AS protein, score
ORDER BY score DESC
LIMIT 5
```

Returned Result:

protein	score
UBC	0.28501
ETS1	0.12924
AR	0.1027
GATA2	0.10168
CTNNB1	0.09567

5 row(s) · 640 ms

35. Query: Betweenness centrality — bridge proteins (sampled, top 5)

Cypher:

```
CALL gds.betweenness.stream('ppi', {samplingSize:1000, samplingSeed:42})
YIELD nodeId, score
RETURN gds.util.asNode(nodeId).node_name AS protein, score
ORDER BY score DESC
LIMIT 5
```

Returned Result:

protein	score
UBC	2030097.25633
CTCF	206518.42825
GATA2	155684.85341
ETS1	143344.21506
APP	131256.73375

5 row(s) · 12644 ms

36. Query: Closeness centrality — proteins central by short distance to all others (top 5)

Cypher:

```
CALL gds.closeness.stream('ppi')
YIELD nodeId, score
RETURN gds.util.asNode(nodeId).node_name AS protein, score
ORDER BY score DESC
LIMIT 5
```

Returned Result:

protein	score
RNU2-1	1
PSPN	1
RNU1-1	1
PRAMEF5	1
TRGC2	1

5 row(s) · 3934 ms

7. GDS Community Detection

Partition the PPI network into communities / modules and measure local clustering. Communities surface functional protein modules.

37. Query: Louvain modularity communities — size of the 5 largest

Cypher:

```
CALL gds.louvain.stream('ppi')
YIELD nodeId, communityId
RETURN communityId, count(*) AS members
ORDER BY members DESC
LIMIT 5
```

Returned Result:

communityId	members
9150	3230
19075	3191
2896	3043

communityId	members
17361	2893
399	2127

5 row(s) · 7330 ms

38. Query: Weakly Connected Components — size of the 5 largest components

Cipher:

```
CALL gds.wcc.stream('ppi')
YIELD nodeId, componentId
RETURN componentId, count(*) AS members
ORDER BY members DESC
LIMIT 5
```

Returned Result:

componentId	members
0	18345
13259	3
18250	2
3697	2
20466	2

5 row(s) · 81 ms

39. Query: Label Propagation communities — 5 largest

Cipher:

```
CALL gds.labelPropagation.stream('ppi', {maxIterations:10})
YIELD nodeId, communityId
RETURN communityId, count(*) AS members
ORDER BY members DESC
LIMIT 5
```

Returned Result:

communityId	members
3	18338
470	7
15019	3
5457	2
110057	2

5 row(s) · 780 ms

40. Query: Modularity Optimization — number of communities and largest size

Cipher:

```
CALL gds.modularityOptimization.stream('ppi', {maxIterations:10})
YIELD nodeId, communityId
RETURN count(DISTINCT communityId) AS communities, count(*) AS nodes_assigned
```

Returned Result:

communities	nodes_assigned
9576	27671

1 row(s) · 5665 ms

41. Query: K-1 Coloring — number of colors needed (graph coloring)

Cypher:

```
CALL gds.k1coloring.stream('ppi')
YIELD nodeId, color
RETURN count(DISTINCT color) AS distinct_colors, count(*) AS nodes
```

Returned Result:

distinct_colors	nodes
44	27671

1 row(s) · 84 ms

42. Query: Triangle Count — proteins forming the most triangles (top 5)

Cypher:

```
CALL gds.triangleCount.stream('ppi')
YIELD nodeId, triangleCount
RETURN gds.util.asNode(nodeId).node_name AS protein, triangleCount
ORDER BY triangleCount DESC
LIMIT 5
```

Returned Result:

protein	triangleCount
UBC	103223
ETS1	23932
CTNNB1	17151
AR	16095
GATA2	14437

5 row(s) · 237 ms

43. Query: Local Clustering Coefficient — graph-wide average cohesion

Cypher:

```
CALL gds.localClusteringCoefficient.stream('ppi')
YIELD nodeId, localClusteringCoefficient
RETURN avg(localClusteringCoefficient) AS avg_clustering_coefficient
```

Returned Result:

avg_clustering_coefficient
0.07523

1 row(s) · 265 ms

8. GDS Similarity & Node Embeddings

Jaccard node similarity over shared interactors, FastRP structural embeddings, and KNN over those embeddings. Embeddings are then persisted to power the vector index in §9.

44. Query: Node Similarity (Jaccard over shared interactors) — top 5 protein pairs

Cipher:

```
CALL gds.nodeSimilarity.stream('ppi', {topK:1, similarityCutoff:0.5})
YIELD node1, node2, similarity
RETURN gds.util.asNode(node1).node_name AS protein_a, gds.util.asNode(node2).node_name
       AS protein_b, similarity
ORDER BY similarity DESC
LIMIT 5
```

Returned Result:

protein_a	protein_b	similarity
TTC22	CCDC175	1
PMS2P4	STAG3L4	1
KRT12	DDT	1
AADAC	SLC7A10	1
OCEL1	CSAG1	1

5 row(s) · 47641 ms

45. Query: FastRP — generate 64-dim structural embeddings (sample vector)

Cipher:

```
CALL gds.fastRP.stream('ppi', {embeddingDimension:64, randomSeed:42})
YIELD nodeId, embedding
RETURN gds.util.asNode(nodeId).node_name AS protein, embedding
LIMIT 3
```

Returned Result:

protein	embedding
PHYHIP	[0.177, -0.064, 0.069, -0.233, ...] (dim=64)
GPANK1	[0.057, -0.028, -0.334, 0.127, ...] (dim=64)
ZRSR2	[-0.206, -0.039, -0.242, -0.265, ...] (dim=64)

3 row(s) · 313 ms

46. Query: FastRP mutate — write 64-dim embeddings into the in-memory projection

Cipher:

```
CALL gds.fastRP.mutate('ppi', {embeddingDimension:64, mutateProperty:'embedding',
                                randomSeed:42})
YIELD nodePropertiesWritten, computeMillis
RETURN nodePropertiesWritten, computeMillis
```

Returned Result:

nodePropertiesWritten	computeMillis
27671	294

1 row(s) · 357 ms

47. Query: KNN over FastRP embeddings (cosine) — 5 nearest-neighbour protein pairs

Cypher:

```
CALL gds.knn.stream('ppi', {nodeProperties:['embedding'], topK:1, randomSeed:42,
    concurrency:1, sampleRate:0.5})
YIELD node1, node2, similarity
RETURN gds.util.asNode(node1).node_name AS protein_a, gds.util.asNode(node2).node_name
    AS protein_b, similarity
ORDER BY similarity DESC
LIMIT 5
```

Returned Result:

protein_a	protein_b	similarity
FAM71E1	DMRTC1	1
PMS2P4	STAG3L4	1
C3orf14	OTOA	1
TTC22	LY6G5C	1
LINC01558	NSUN5P2	1

5 row(s) · 10163 ms

48. Query: Persist FastRP embeddings from projection back to Neo4j nodes

Cypher:

```
CALL gds.graph.nodeProperties.write('ppi', ['embedding'])
YIELD propertiesWritten
RETURN propertiesWritten
```

Returned Result:

propertiesWritten
27671

1 row(s) · 380 ms

49. Query: List in-memory GDS graphs (catalog inventory)

Cypher:

```
CALL gds.graph.list()
YIELD graphName, nodeCount, relationshipCount, memoryUsage
RETURN graphName, nodeCount, relationshipCount, memoryUsage
```

Returned Result:

graphName	nodeCount	relationshipCount	memoryUsage
ppi	27671	642150	15618 KiB

1 row(s) · 61 ms

9. Indexes, Constraints, Full-text & Vector Search

Schema objects that make production queries fast: range/text indexes, uniqueness constraints, a full-text index over entity names, and a cosine VECTOR index over the FastRP embeddings written in §8.

50. Query: Create a RANGE index on gene_protein.node_name

Cypher:

```
CREATE RANGE INDEX gene_name_idx IF NOT EXISTS FOR (n:gene_protein) ON (n.node_name)
```

Returned Result:

(no rows returned) · 295 ms

51. Query: Create a TEXT index on disease.node_name (substring search acceleration)

Cypher:

```
CREATE TEXT INDEX disease_name_text IF NOT EXISTS FOR (n:disease) ON (n.node_name)
```

Returned Result:

(no rows returned) · 8 ms

52. Query: Create a uniqueness CONSTRAINT on drug.node_index

Cypher:

```
CREATE CONSTRAINT drug_idx_unique IF NOT EXISTS FOR (d:drug) REQUIRE d.node_index IS  
UNIQUE
```

Returned Result:

(no rows returned) · 20 ms

53. Query: List all indexes currently defined

Cypher:

```
SHOW INDEXES  
YIELD name, type, entityType, labelsOrTypes, properties  
RETURN name, type, labelsOrTypes, properties  
ORDER BY name
```

Returned Result:

name	type	labelsOrTypes	properties
disease_name_text	TEXT	[disease]	[node_name]
drug_idx_unique	RANGE	[drug]	[node_index]
entity_names	FULLTEXT	[drug, disease, gene_protein]	[node_name]
gene_name_idx	RANGE	[gene_protein]	[node_name]
index_343aff4e	LOOKUP	None	None
index_f7700477	LOOKUP	None	None
protein_embedding_vec	VECTOR	[gene_protein]	[embedding]

7 row(s) · 54 ms

54. Query: List all constraints currently defined

Cypher:

```
SHOW CONSTRAINTS  
YIELD name, type, labelsOrTypes, properties  
RETURN name, type, labelsOrTypes, properties
```

Returned Result:

name	type	labelsOrTypes	properties
drug_idx_unique	UNIQUENESS	[drug]	[node_index]

1 row(s) · 34 ms

55. Query: Create a FULLTEXT index over names of drugs, diseases and genes

Cypher:

```
CREATE FULLTEXT INDEX entity_names IF NOT EXISTS FOR (n:drug|disease|gene_protein) ON
EACH [n.node_name]
```

Returned Result:

(no rows returned) · 11 ms

56. Query: Full-text search for 'anemia' across all entity names (ranked)

Cypher:

```
CALL db.index.fulltext.queryNodes('entity_names', 'anemia')
YIELD node, score
RETURN labels(node)[0] AS label, node.node_name AS name, score
ORDER BY score DESC
LIMIT 6
```

Returned Result:

label	name	score
disease	anemia (disease)	3.07386
disease	congenital anemia	3.07386
disease	neonatal anemia	3.07386
disease	sideroblastic anemia	3.07386
disease	pernicious anemia	3.07386
disease	refractory anemia	3.07386

6 row(s) · 60 ms

57. Query: Create a cosine VECTOR index over the 64-dim FastRP embeddings

Cypher:

```
CREATE VECTOR INDEX protein_embedding_vec IF NOT EXISTS FOR (n:gene_protein) ON
(n.embedding) OPTIONS {indexConfig: {'vector.dimensions': 64,
'vector.similarity_function': 'cosine'}}
```

Returned Result:

(no rows returned) · 61 ms

58. Query: Vector KNN search — proteins most similar to a query protein's embedding

Cypher:

```
MATCH (q:gene_protein {node_name:$name})
WHERE q.embedding IS NOT NULL
CALL db.index.vector.queryNodes('protein_embedding_vec', 6, q.embedding)
YIELD node, score
RETURN node.node_name AS similar_protein, score
ORDER BY score DESC
```

Returned Result:

similar_protein	score
UBC	0.99987
UBR5	0.94399
PGAM5	0.94314
VAV2	0.94189
PRKN	0.94084
DDT	0.94028

6 row(s) · 35 ms

10. APOC Procedures & Functions

APOC helpers for meta-modelling, weighted degree, path expansion, collection maths and text utilities.

59. Query: APOC node degree (total) for a hub protein

Cypher:

```
MATCH (n:gene_protein {node_name:$name})
RETURN apoc.node.degree(n) AS total_degree
```

Returned Result:

total_degree
5580

1 row(s) · 8 ms

60. Query: APOC meta-graph: which node labels connect via which relationships (sample)

Cypher:

```
CALL apoc.meta.graphSample()
YIELD nodes, relationships
RETURN size(nodes) AS label_count, size(relationships) AS rel_pattern_count
```

Returned Result:

label_count	rel_pattern_count
10	33

1 row(s) · 35 ms

61. Query: APOC path expansion: unique proteins within 2 PPI hops of a hub

Cypher:

```
MATCH (n:gene_protein {node_name:$name})
CALL apoc.path.subgraphNodes(n, {relationshipFilter:'protein_protein', maxLevel:2})
YIELD node
RETURN count(node) AS nodes_within_2_hops
```

Returned Result:

nodes_within_2_hops
14651

1 row(s) · 86 ms

62. Query: APOC collection stats over a degree sample

Cypher:

```
MATCH (n:gene_protein)-[:protein_protein]-()
WITH n, count(*) AS d
LIMIT 500
WITH collect(d) AS degs
RETURN apoc.coll.max(degs) AS max, apoc.coll.min(degs) AS min, apoc.coll.avg(degs) AS avg
```

Returned Result:

max	min	avg
5197	2	157.378

1 row(s) · 35 ms

63. Query: APOC text utilities: clean & capitalize a drug name

Cypher:

```
RETURN apoc.text.capitalizeAll(toLower($drug)) AS pretty_name
```

Returned Result:

pretty_name

Copper

1 row(s) · 9 ms

11. Production Analytical Scenarios

End-to-end questions a biomedical analyst or the Eugene chat agent would actually ask — drug repurposing, hub discovery, shared-target reasoning and disease comorbidity structure.

64. Query: Drug-repurposing: drugs targeting proteins linked to a disease

Cypher:

```
MATCH (dis:disease)-[:disease_protein]-(p:gene_protein)-[:drug_protein]-(d:drug)
WHERE dis.node_name CONTAINS 'anemia'
RETURN d.node_name AS candidate_drug, count(DISTINCT p) AS shared_targets
ORDER BY shared_targets DESC
LIMIT 6
```

Returned Result:

candidate_drug	shared_targets
Copper	10
Zinc acetate	8
Zinc sulfate, unspecified form	8
Zinc	8
Zinc chloride	8
Arteminol	8

6 row(s) · 21 ms

65. Query: Most disease-associated proteins (potential master regulators)

Cypher:

```
MATCH (p:gene_protein)-[:disease_protein]-(d:disease)
RETURN p.node_name AS protein, count(DISTINCT d) AS disease_count
ORDER BY disease_count DESC
LIMIT 6
```

Returned Result:

protein	disease_count
SOD2	273
TP53	257
PTEN	247
TNF	225
IL6	208
KRAS	200

6 row(s) · 168 ms

66. Query: Shared targets between two drugs (mechanistic overlap)

Cypher:

```
MATCH (d1:drug)-[:drug_protein]-(p:gene_protein)-[:drug_protein]-(d2:drug)
WHERE d1.node_name < d2.node_name
RETURN d1.node_name AS drug_a, d2.node_name AS drug_b, count(DISTINCT p) AS shared
ORDER BY shared DESC
LIMIT 6
```

Returned Result:

drug_a	drug_b	shared
Zinc chloride	Zinc sulfate, unspecified form	126
Zinc	Zinc acetate	125
Zinc acetate	Zinc sulfate, unspecified form	93
Zinc acetate	Zinc chloride	93
Zinc	Zinc chloride	90
Zinc	Zinc sulfate, unspecified form	90

6 row(s) · 6593 ms

67. Query: Disease comorbidity by shared phenotypes (anemia cohort, top pairs)

Cypher:

```
MATCH (a:disease)
WHERE a.node_name CONTAINS 'anemia'
MATCH
    (a)-[:disease_phenotype_positive]-(ph:effect_phenotype)-[:disease_phenotype_positive]-(b:disease)
WHERE a.node_name < b.node_name
WITH a, b, count(DISTINCT ph) AS shared_phenotypes
RETURN a.node_name AS disease_a, b.node_name AS disease_b, shared_phenotypes
ORDER BY shared_phenotypes DESC
LIMIT 6
```

Returned Result:

disease_a	disease_b	shared_phenotypes
Fanconi anemia complementation group	X-linked intellectual disability	58
Fanconi anemia complementation group	developmental and epileptic encephalopathy	49
Fanconi anemia complementation group	orofaciodigital syndrome	48
Fanconi anemia complementation group	intellectual disability, autosomal dominant	46
Diamond-Blackfan anemia	X-linked intellectual disability	45
Diamond-Blackfan anemia	Fanconi anemia complementation group	44

6 row(s) · 720 ms

68. Query: Pathway enrichment: pathways with the most member proteins

Cypher:

```
MATCH (pw:pathway)-[:pathway_protein]-(p:gene_protein)
RETURN pw.node_name AS pathway, count(DISTINCT p) AS proteins
ORDER BY proteins DESC
LIMIT 6
```

Returned Result:

pathway	proteins
Neutrophil degranulation	478
Olfactory Signaling Pathway	393
Generic Transcription Pathway	346
Antigen processing: Ubiquitination & Proteasome degradation	309

pathway	proteins
G alpha (i) signalling events	259
Neddylation	234

6 row(s) · 47 ms